Glance ML Internship Assignment

Multimodal Fashion & Context Retrieval

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1. Project Overview

The goal of this assignment is to build an intelligent search engine that can retrieve specific images from a diverse database based on natural language descriptions. The system needs to understand not just "what" someone is wearing, but "where" they are and the "vibe" of their attire.

2. The Dataset

You are responsible for sourcing or simulating a dataset (minimum 500-1,000 images). The data must contain variations across three primary axes:

- Environment: Office interiors, urban streets, parks, and home settings.
- Clothing Types: Formal (blazers, button-downs), Casual (hoodies, t-shirts), and Outerwear, etc
- Color Theory: A wide palette of garment colors.

One such dataset is fashionpedia's <u>dataset or you can download it here</u>

3. Technical Requirements

Your submission must consist of two distinct workflows, ideally hosted in 1 GitHub repository in different directories or files, clearly defined modules:

Part A: The Indexer

- Feature Extraction: Process the raw images into a searchable format.
- Vector Storage: Implement a solution to store these representations efficiently (avoiding simple filename keyword matching).

Part B: The Retriever (The "Query")

- Search Logic: Create a script that accepts a natural language string and returns the top k matching images.
- Context Awareness: The system should handle multi-attribute queries (e.g., color + clothing type + location).

Focus more on the ML logic aspect of it, rather than the engg work of indexing, for example - if you chose to pick a Vector DB, pick the easiest and most convenient one instead of spending time and code on rewriting your own version. You are assessed first and foremost on ML logic.

Hint: While architectures like CLIP provide a strong baseline for zero-shot retrieval, they often struggle with compositionality (e.g., distinguishing "red shirt with blue pants" from "blue shirt with red pants") and fine-grained fashion attributes. The expected solution should be better than vanilla application of CLIP with focus on making it work for fashion based retrieval.

4. Evaluation Queries

Your system will be judged on its ability to accurately return images for the following prompts:

1. Attribute Specific: "A person in a bright yellow raincoat."

2. Contextual/Place: "Professional business attire inside a modern office."

3. Complex Semantic: "Someone wearing a blue shirt sitting on a park bench."

4. Style Inference: "Casual weekend outfit for a city walk."

5. Compositional: "A red tie and a white shirt in a formal setting."

5. Submission Deliverables

A single PDF that contains

1. Approaches: Possible ways to solve this problem, tradeoffs, what's good and when

2. Short Write-up on Chosen Approach: A brief explanation of your chosen architecture & how it handles fashion queries

3. Codebase (GitHub) Link: Clean, documented code for both the indexing and retrieval pipelines.

4. Approaches for future work

  - a. How to extend this solution for adding locations (cities, places) and weather

  - b. How to improve precision

6. What We Are Looking For

1. Thoughtful solution:

  - a. Getting a solution in the age of AI is easy - coding it successfully, understanding the chosen approach (model, arch), what are its shortcomings & how to address that are sought

  - b. How well does it work for fashion based queries?

2. Modular Code: Is your logic separated from your data?

3. Scalability: Would your retrieval logic work if the dataset grew to 1 million images?

4. Zero-Shot Capability: How well does the system handle descriptions it hasn't seen explicitly in a

training label?